

NDI DEFINITION SET · DOCUMENT 2A (COMPANION TO DOCUMENT 2)

The Feynman Grand Prize as a Verifiable Task Tree

A capability-pinned restatement of the prize, decomposed into machine-verifiable tasks — from feedstock-level reaction sequences to arm subsystems — across two contrasting construction chemistries

Version 0.1 · 2026-06-11 · for funding & strategic partners

Builds directly on Document 2 (the eight capability axes and the per-capability acceptance protocol). It does not alter the Foresight prize; it makes that prize precise enough to be decomposed and adjudicated by automated and instrumental verifiers.

Purpose. Document 2 redefined molecular nanotechnology as eight measurable capability axes. Here we apply that frame to the prize itself: we (1) pin each of the two devices to numeric capability targets, (2) define a machine-verifiable task — a node carrying a spec, numeric acceptance criteria, and a named verifier (DFT/MD simulation, metrology instrument, or exhaustive logic check) — and (3) decompose the prize top-to-bottom into a tree of such tasks. We work the construction chemistry two ways: the Freitas-Merkle minimal diamond-mechanosynthesis toolset¹ as the canonical positional-covalent route, and a positionally-controlled single-site-catalysis route as its material-agnostic contrast. We then decompose the arm into its subsystems — struts, joints, actuation, power transmission, sensing, and the charge-delivery wiring that drives the chemistry — and the adder into its logic, interconnect, output, and energy budget. The whole tree is also ingested into the the knowledge graph knowledge graph as a hierarchical workflow of verifiable claims.

Contents

1. The prize, pinned to capability targets

The original prize names two devices. We restate each as a vector of numeric targets on the eight axes of Document 2. Where the 1995 text is explicit we keep its number; where it was silent (force, energy, environment, verifiability) we supply the capability-extension value. This table is the root specification from which every downstream task inherits its acceptance threshold.

Axis	Device A — builder/arm	Device B — adder	Source / note
Positional precision	≤ 0.1 nm RMS placement (0.03 nm verify floor)	n/a (logic device)	1995 literal; verify floor = $\frac{1}{2}$ rule (Doc 2)
Programmability	Build judge-specified target revealed at test time	Add any of all 65,536 input pairs	1995 implied; made explicit
Atomic precision	≤ 1 defect / 10^4 – 10^6 placements	100% logical correctness	Capability extension / 1995 literal
Delivered force	Source the force the chosen chemistry needs at site, under load	n/a	Capability extension (the 1995 omission)
Elemental generality	$\geq N$ element classes placed & bonded	n/a	Capability extension (figure of merit)
Product stiffness	Stiff product where claimed (to $\sim 10^2$ – 10^3 GPa)	n/a	Capability extension
Throughput / duty	$\geq 1,000$ motions/s (~ 500 rxns/s) for ≥ 60 s	Speed unconstrained	1995 literal → reaction-rate restatement
Energy	(actuation budget stated)	≤ 100 k_BT per bit-operation	Capability extension (the adder's missing FoM)
Geometry	Fits 100 nm cube; works in 50 nm cube	Fits 50 nm cube	1995 literal
Verifiability	Every claim independently confirmable	All inputs externally applied; output read without entrant interpretation	Capability extension (Doc 2 §3.2)
Statistics	32 copies, destructive test	32 copies, destructive test	1995 literal (retained)

Table 1. The capability-pinned prize specification. *Italicised axes (force, energy, generality, stiffness, verifiability) are the additions that make the prize objectively scorable; the remainder are the 1995 literals.*

Environment is not neutral and must be declared per route: the diamond-mechanosynthesis route is specified for ultra-high vacuum and cryogenic operation (**80 K nominal**, moderate reliability at 300 K),¹ whereas catalytic and scaffold routes may run at ambient or in liquid. We treat operating environment as a stated input to each task, not a global constant.

2. What a machine-verifiable task is

A task is the atomic unit of the decomposition. It is **machine-verifiable** when an automated or instrumental procedure — not a human judgement — can return pass/fail against numeric criteria. Every task node in this document carries the following fields:

Field	Meaning
Goal	The single capability outcome the task establishes.
Inputs → Outputs	Feedstock/parts/signals consumed; the artifact or measurement produced.
Preconditions	Tasks or states that must hold first (the dependency edges of the tree).
Procedure	The operation(s) performed — a reaction sequence, an actuation, a measurement.

Field	Meaning
Acceptance criteria	Numeric thresholds that define success (energies, forces, error rates, tolerances).
Verifier	The named adjudicator: DFT/MD simulation, a metrology instrument, or an exhaustive logic check — tied to Document 2 Table 4.
Failure modes	Known pathologies the verifier must screen for and the bound on each.
Axes	Which of the eight capability axes the task advances.

Table 2. The task schema. Each leaf in the tree is one instance of this record.

Three verifier classes. (i) **In-silico** — DFT or MD on the tool/workpiece complex returns barrier heights, reaction energies, and pathology branching ratios; this is the verifier the Freitas–Merkle toolset already supplies for 1,620 structures.¹ (ii) **Instrumental** — the metrology of Document 2 §3.2 (AC-STEM ptychography, nc-AFM, APT, force spectroscopy) measures the physical product against tolerance, with measurement uncertainty $\leq \frac{1}{3}$ of the claim. (iii) **Logical** — exhaustive enumeration (all 65,536 adder cases) or continuity/duty logging. A task is only admitted to the tree once its verifier and threshold are named.

Mapping to the graph. The tree is isomorphic to a hierarchical workflow: the root thesis is the capability-pinned prize (Table 1); phases are the decomposition branches (§4–§9); steps are the task cards; and dependency edges become the workflow's prerequisite relations. Section 10 records the ingest.

3. The decomposition at a glance

The prize splits into two devices; the builder splits into a **construction-chemistry** stack and a **positional-machine** stack; the positional machine splits into structure, motion, power, sensing, and charge delivery. The chemistry stack is worked twice — once per route — because the route chosen sets the force, feedstock, and environment every subsystem below must serve.

Branch	Decomposes into
PRIZE	Device A (builder + arm) Device B (adder)
A · Construction chemistry	Route A1: positional diamond mechanosynthesis (§4) Route A2: single-site catalytic construction (§5)
A · Positional machine	Struts (§6.1) · Joints/bearings (§6.2) · Actuation (§6.3) · Power transmission (§6.4) · Sensing & feedback (§6.5) · Charge delivery & wiring (§6.6) · Tooltip interface & recharge (§6.7) · Whole-arm integration (§6.8)
B · Molecular adder	Logic primitive · Interconnect · Output transducer · Energy accounting (§9)

Table 3. Top of the task tree. Each leaf below is rendered as a task card carrying the Table 2 fields.

4. Construction route A1 — positional diamond mechanosynthesis

The Freitas–Merkle *minimal toolset*¹ is the reference positional-covalent route and, for our purposes, the gold standard of a **pre-verified** task set: it already specifies, with DFT, every reaction step and side reaction. The toolset uses **three primary tools** — Hydrogen Abstraction (HAbst), Hydrogen Donation (HDon), Dimer Placement (DimerP) — and **six auxiliary tools** — AdamRad and GeRad handles, Methylene (Meth), Germylmethylene (GM), Germylene (Germ), and Hydrogen Transfer (HTrans). It draws on three elements (C, Ge, H) from four bulk feedstocks (CH₄, C₂H₂, Ge₂H₆, H₂), and exhibits 100% process closure: it can fabricate all nine tools, recharge them after use, and build clean and hydrogenated C(111)/C(110)/C(100) and hexagonal diamond of unlimited size, plus graphene and nanotubes.¹

The published analysis compiled **65 reaction sequences, 328 reaction steps, 354 unique pathological side reactions, and 1,321 DFT energies** over 1,620 tool/workpiece structures at the B3LYP/6-311+G(2d,p)//B3LYP/3-21G* level. Each reaction sequence is, in our terms, a ready-made machine-verifiable task: the DFT energy and barrier are the verifier, and the catalogued pathologies are the failure-mode bounds. The reliability target is high at 80 K, moderate at 300 K.¹

4.1 The canonical worked task — place one carbon dimer

A1-CORE — Place one C ₂ dimer on a dehydrogenated C(110) site (DCB6Ge cycle)	
Goal	Add two carbon atoms at a judge-specified lattice site to ≤0.1 nm, then recharge the tool — the irreducible growth step of covalent diamond construction.
Inputs → Outputs	DimerP tool charged with a C ₂ dimer (from C ₂ H ₂) + prepared radical site → two C atoms bonded into the lattice; spent tool.
Preconditions	Site dehydrogenated by HAbst (A1-HAbst); adjacent geometry within steric/force window; UHV, ~80 K.
Procedure	Position DimerP over site; approach along the validated low-barrier trajectory; transfer C ₂ ; retract leaving the dimer bonded (DCB6Ge places C ₂ onto C(110)).
Acceptance	Reaction exothermic along the modelled path; transfer barrier below the tabulated ceiling (improves to ~+0.10 eV on the optimal trajectory); placed-dimer geometry within 0.1 nm of target; no dimer retained on tool. Verify floor ≤0.03 nm.
Verifier	In-silico: DFT (B3LYP) on the tool/workpiece complex for energy + barrier. Instrumental: nc-AFM / AC-STEM confirms placement on the built artifact.
Failure modes	Dimer not deposited (tool over-binds); mis-site placement; <i>ethenylation</i> and other side reactions — each held below the toolset's catalogued branching ratio.
Axes	Positional precision · Delivered force · Atomic precision · Throughput

4.2 The supporting reaction-sequence tasks

The dimer task depends on a small set of equally-verifiable sub-tasks drawn directly from the toolset's reaction sequences. Each is a node with the same schema; thresholds are the published DFT energies and pathology bounds.

Task	Goal	Verifier / threshold	Failure bound
A1-HAbst	Abstract one H to create a radical site	DFT: abstraction exothermic, barrier below ceiling; site is monoradical	No double-abstraction; tool keeps H
A1-HDon	Donate one H to passivate / cap a site	DFT: donation favourable; correct valence	No mis-donation to wrong site
A1-Recharge	Recharge each of the 9 tooltypes after use	DFT closure: spent tool +	Recharge dead-ends (closure

Task	Goal	Verifier / threshold	Failure bound
		feedstock → charged tool	< 100%)
A1-Fab	Fabricate each tool from adamantane seed / flat surface	DFT build sequence completes	Intermediate not reachable
A1-Pathology	Screen each step against catalogued side reactions	Branching ratio ≤ tabulated bound at 80 K	Pathology dominates at 300 K

Table 4. Route-A1 reaction-sequence tasks. All five are pre-verified in the published toolset; the contribution here is to cast them in the common task schema so they sit in the same tree as the engineering subsystems and the catalytic route. (65 sequences / 328 steps total available.)¹

5. Construction route A2 — single-site catalytic crystal construction

Route A2 targets the **same deliverable** — an atomically precise, stiff crystal at a judge-specified site, in diamond or another material (SiC, a compound semiconductor, an oxide) — but replaces mechanically-forced covalent addition with a **positionally-controlled single-site transition-metal catalyst** that lowers the reaction barrier rather than forcing through it. The motivation, from Document 2, is twofold: it cuts the delivered-force requirement from the ~6 nN of an uncatalyzed Ge–C addition toward the ~100 pN–nN range, and because catalysis spans much of the periodic table it opens elemental generality that the C/Ge/H toolset does not reach. The cost is maturity: where Route A1 is DFT-closed, Route A2's force-reduction and selectivity-under-load are **hypotheses the tasks below are designed to verify**, not established results.

5.1 The catalytic worked task

A2-CORE — Add one growth unit at a specified site by single-site catalysis, under load	
Goal	Place and bond one growth unit (e.g. a C or Si addition, or an oxide formula unit) at a judge-specified site via a positioned single-site catalyst, holding site selectivity while the tool bears mechanical load.
Inputs → Outputs	Catalyst-functionalised tip + molecular precursor → one growth unit incorporated; catalyst regenerated.
Preconditions	Catalyst attached and characterised; precursor delivery; environment per chosen chemistry (may be ambient/liquid).
Procedure	Position catalyst over site; admit precursor; catalytic turnover bonds the unit; verify regeneration for the next cycle.
Acceptance	Delivered force at site ≤ 1 nN (target ~100 pN–nN) measured during a real placement; site selectivity retained under that load; turnover confirmed; catalyst regenerated; defect rate ≤ route target.
Verifier	In-silico: DFT of the catalytic cycle (barrier, selectivity, regeneration). Instrumental: AFM force spectroscopy for the force-under-load claim; EELS/APT for placed-species identity. (This is the load-bearing verification of the whole catalytic thesis.)
Failure modes	Loss of selectivity under load; catalyst poisoning / non-regeneration; force floor not actually reduced; wrong-species incorporation.
Axes	Delivered force · Elemental generality · Atomic precision · Positional precision

5.2 Route A1 vs A2 — the contrast on the schema

Casting both routes in the same task schema makes the trade explicit: identical deliverable and acceptance geometry, opposite positions on force, feedstock generality, environment, and verification maturity.

Schema field	A1 · diamond mechanosynthesis	A2 · single-site catalysis
Mechanism	Mechanically force reactant through barrier	Catalyst lowers barrier; position selects site
Delivered force	~6 nN (Ge–C); high stiffness mandatory	~100 pN–nN target; modest stiffness (claim to verify)
Feedstock / elements	CH ₄ , C ₂ H ₂ , Ge ₂ H ₆ , H ₂ → C, Ge, H	Precursor + catalyst → much of the periodic table
Product	Diamond / graphene / nanotube	Diamond or SiC, oxides, compound semiconductors
Environment	UHV, ~80 K	Often ambient or liquid
Verifier maturity	DFT-closed: 65 seq / 1,321 energies	Open: cycle + force-under-load must be demonstrated
Chief risk	Force/stiffness engineering of the arm	Selectivity & force-reduction may not hold under load

Table 5. The two routes are complements, not rivals: A1 is the proven-on-paper hard path; A2 is the higher-generality, lower-force bet whose verifiers are the point of the program.

6. The positional machine — subsystem task tree

Whichever chemistry is chosen, the builder needs a machine that positions the tooltip and supplies what the reaction demands. We decompose it into eight subsystems. The acceptance criteria below inherit from Table 1 and from the chosen route's force figure: A1 forces every stiffness number to its hard limit; A2 relaxes the force-driven ones. Numbers marked *TBR* (to be ratified) are design targets pending the route decision.

6.1 Structural members (struts / links)

S1 — Load-path struts that hold the tooltip frame rigid under reaction force	
Goal	Carry the reaction-force load path from anchor to tooltip with deflection $\ll$ placement tolerance.
Acceptance	End-to-end tip stiffness $k \geq 60 \text{ N/m}$ for the A1 (6 nN) route so deflection under load $< 0.1 \text{ nm}$; $\geq \sim 1\text{--}3 \text{ N/m}$ may suffice for A2. Stiff member material to $\sim 1 \text{ TPa}$ (diamondoid) class. (TBR per route.)
Verifier	FEM/MD stiffness model + measured resonance/compliance on the built member; deflection-under-rated-load test.
Failure modes	Static deflection $>$ tolerance (the original prize's silent failure); buckling; creep.
Axes	Delivered force · Positional precision · Product stiffness

6.2 Joints & bearings

S2 — Low-backlash, stiff articulations spanning the workspace	
Goal	Provide the degrees of freedom to reach any point in the 50 nm working cube while preserving stiffness and repeatability.
Acceptance	Backlash / play $\leq 0.03 \text{ nm}$ ($\frac{1}{3}$ of tolerance); joint stiffness consistent with the S1 load path; full reach of the 50 nm cube; repeatable return-to-position.
Verifier	Kinematic model + metrology of commanded-vs-actual pose (AC-STEM/nc-AFM fiducials); hysteresis loop measurement.
Failure modes	Backlash, hysteresis, stiction; stiffness loss at extended reach; wear over duty cycle.
Axes	Positional precision · Programmability · Throughput

6.3 Actuation

The actuator converts an off-board command into tip motion and, where the chemistry needs it, into reaction force. Candidate physics differ sharply in force density, stroke, and bandwidth; the choice is route-dependent.

Actuator class	Force / stroke	Bandwidth	Fit
Electrostatic	Modest force, nm stroke	High ($\geq 1 \text{ kHz}$)	Fine positioning; A2 forces
Piezoelectric	High force, sub-nm–nm	High	Fine positioning both routes
Thermal / bimorph	High force, larger stroke	Low (slow)	Coarse stage only
Chemo-mechanical	Variable	Low	Speculative; not for duty rate

Table 6. Actuator options. A1's $\sim 6 \text{ nN}$ reaction force plus the $\geq 1,000$ motions/s duty rate is the binding constraint; A2 relaxes the force but keeps the rate.

S3 — Actuator delivering positioning stroke, reaction force, and duty rate	
Goal	Move the tip across the workspace and source the reaction force at $\geq 1,000$ motions/s for $\geq 60 \text{ s}$.
Acceptance	Force capability $\geq$ route requirement ($\approx 6 \text{ nN}$ A1 / $\sim 100 \text{ pN}$ – nN A2); positioning resolution $\leq 0.03 \text{ nm}$; $\geq 1,000$ motions/s sustained $\geq 60 \text{ s}$; stated actuation energy budget.
Verifier	Force-displacement characterisation; continuous duty log (no stitching); energy-per-motion

	accounting.
Failure modes	Force sag under load; bandwidth/stroke trade failure; thermal drift; energy budget blown.
Axes	Delivered force · Throughput · Positional precision · Energy

6.4 Power transmission

Power and motion must reach the active tip from off-board. The transmission medium is a first-class design fork — electrical lines, stiff mechanical push-rods (rod logic), pneumatic/hydraulic channels, or Bowden/tendon (tension) cables — each with different stiffness, latency, and routing cost across the moving joints.

Medium	Carries	Strength	Weakness
Electrical wiring	Charge, signals, e-static drive	Fast; light; co-routes with sensing	Leakage; insulation; field crosstalk
Mechanical push-rod	Position / force (rod logic)	Very stiff; deterministic	Routing bulk; limited DOF; backlash
Pneumatic / hydraulic	Force / pressure	High force density	Compliant fluid; sealing at scale
Bowden / tendon cable	Tension actuation	Routes around joints; light	Tension-only; friction; compliance

Table 7. Power-transmission media. Tension (Bowden/tendon) and stiff push-rods trade routability against stiffness; electrical co-routes with the charge-delivery subsystem (§6.6).

S4 — Transmission path delivering power/motion to the tip without spoiling stiffness	
Goal	Convey actuation power and/or motion from anchor to tip across all joints within the stiffness and latency budget.
Acceptance	Transmission compliance does not drop tip stiffness below S1; command latency within the duty-rate period ($\leq \sim 1$ ms for 1 kHz); routing fits the 100 nm envelope.
Verifier	End-to-end stiffness/latency model + measured step response at the tip; envelope/packing check.
Failure modes	Cable stretch / rod backlash / fluid compliance; friction losses; routing collisions; latency > period.
Axes	Delivered force · Throughput · Positional precision

6.5 Sensing & feedback

Position can be feedback-stabilised on a compliant structure; static reaction force cannot — so the sensing suite must report **four** channels independently: tip position, contact force, drive current, and local field. Each is a verifiable measurement task with its own resolution floor.

Channel	What it closes	Resolution target	Method
Position	Placement tolerance loop	≤ 0.03 nm	Tunnel-current / interferometric / fiducial
Force	Reaction-force & selectivity-under-load	~ 10 pN class	AFM force spectroscopy / resonance shift
Current	Charge driving the chemistry	sub-pA	Transimpedance readout
Field	Local electrostatic state at tip	mV / ($\text{V}\cdot\text{nm}^{-1}$) class	Kelvin-probe / field sensing

Table 8. The four feedback channels. Position and force together are what the 1995 spec conflated; current and field are needed only for charge-driven chemistries (§6.6).

S5 — Four-channel feedback meeting the 1/3-uncertainty rule on every channel	
Goal	Measure position, force, current, and field at the tip with uncertainty $\leq 1/3$ of the controlled

	tolerance, in real time.
Acceptance	Each channel meets its Table 8 floor; loop bandwidth $\geq$ actuation bandwidth; force channel proves selectivity is held under load (not feedback-reconstructed).
Verifier	Calibration against reference standards; closed-loop step-response; cross-check of force channel by independent spectroscopy.
Failure modes	Position/force crosstalk; insufficient bandwidth; drift; faking force via position feedback.
Axes	Verifiability · Positional precision · Delivered force

6.6 Charge delivery & wiring

If the chemistry is electrically driven — applying a voltage or injecting charge to open a reaction (as in STM depassivation and many catalytic cycles) — the arm must distribute that charge to the active site and confine it there. This subsystem is the wiring, insulation, and grounding that carry potential through a moving structure without leaking into the load path or the sensing channels.

S6 — Distribute drive voltage/charge to the active tip; confine and account for it	
Goal	Deliver a specified potential / charge to the reaction site through the articulated arm, with controlled leakage and no corruption of force or position sensing.
Inputs → Outputs	Off-board voltage/current source → controlled potential at tip; measured charge delivered per reaction.
Acceptance	Delivered potential within tolerance at the site; leakage current $\leq$ stated budget ($\ll$ drive current) ; charge-per-reaction logged for the \$9 energy account; field crosstalk into S5 below its noise floor.
Verifier	Electrical characterisation (I-V, leakage); field map at tip; correlation check that charge injection does not bias the position/force channels.
Failure modes	Dielectric leakage / breakdown; electromigration in nm wiring; field crosstalk into sensing; unaccounted charge.
Axes	Delivered force (chemistry enablement) · Verifiability · Energy

6.7 Tooltip interface & recharge magazine

S7 — Mount, exchange, and recharge tooltips without losing registration	
Goal	Present the right charged tool to the site, exchange spent tools, and recharge them (the A1 closure requirement) while preserving the position frame.
Acceptance	Tool-exchange repeatability ≤ 0.03 nm; recharge closure 100% of tool types (A1); exchange time within duty budget.
Verifier	Registration metrology before/after exchange; recharge-sequence DFT closure (A1) or catalyst-regeneration assay (A2).
Failure modes	Registration loss on exchange; incomplete recharge (closure $< 100\%$); tool contamination.
Axes	Programmability · Throughput · Atomic precision

6.8 Whole-arm integration

S8 — End-to-end: build a judge-specified object to tolerance, verified	
Goal	Compose S1–S7 plus the chosen chemistry route to construct a test-time-revealed target object and pass independent metrology.
Acceptance	Built object within 0.1 nm of the specified geometry (verify ≤ 0.03 nm); defect rate $\leq$ route target over $\geq 10^4$ placements; $\geq 1,000$ motions/s sustained; 32 copies survive destructive test.

Verifier	Correlative AC-STEM/nc-AFM/APT on the destructive-test population; continuous duty log; second-lab replication for any apex claim.
Failure modes	Subsystem interactions (drift, crosstalk, thermal); statistics fail on full population; non-reproducible.
Axes	All eight

7. Device B — the molecular adder task tree

The adder decomposes into four tasks. Correctness is a logical verifier (exhaustive enumeration); energy is the figure of merit Document 2 restored.

Task	Goal	Verifier / threshold	Failure bound
B1-Logic	8-bit addition primitive in a 50 nm cube	Logic: correct on a covering set in-silico	Logic error on any case
B2-Exhaustive	Correct across all 65,536 input pairs	Logical: full enumeration, externally applied inputs	Any incorrect sum
B3-Output	Unambiguous readable output (any modality)	Instrumental: output read without entrant interpretation	Ambiguous / entrant-decoded output
B4-Energy	≤ 100 k_BT per bit-operation	Calorimetric / electrical dissipation; basis defined	Exceeds ceiling; bit-op undefined

Table 9. Device-B tasks. B4 requires a stated definition of “bit-operation” per 8-bit addition before the ceiling is meaningful (≈140× the Landauer limit).

B-CORE — Compute every 8-bit sum within volume and energy bounds	
Goal	Add any two 8-bit numbers (overflow discarded) inside a 50 nm cube, correctly and within the energy ceiling.
Inputs → Outputs	Two 8-bit inputs (externally applied, any admitted signal class) → readable sum.
Acceptance	100% correct over all 65,536 pairs; fits 50 nm cube; ≤ 100 k_BT/bit-op; output read without entrant interpretation.
Verifier	Logical exhaustion + instrumental output read + energy measurement (Doc 2 §3.2).
Failure modes	Single-case logic error; over-volume; energy over ceiling; ambiguous output.
Axes	Programmability · Energy · Verifiability · Geometry

8. The task tree as a hierarchical workflow

The full tree is represented as a hierarchical workflow so that each task becomes a claim node carrying its acceptance criteria and verifier, and the dependency edges become prerequisite relations. This lets the decomposition be queried, challenged, and updated as evidence arrives (e.g. when the Route-A2 force-under-load verifier returns a real number).

Ingest field	Value
canonical_name	feynman-grand-prize-verifiable-task-tree
thesis	The capability-pinned prize (Table 1), satisfiable by either construction route.
phases	Precise spec · Route A1 · Route A2 · Structure · Joints · Actuation · Power transmission · Sensing · Charge delivery · Tooltip/recharge · Whole-arm · Device B
steps	The task cards (A1-CORE, A2-CORE, S1-S8, B-CORE, ...) with acceptance criteria as operations.
relationships	decomposes_to from the prize thesis to each phase; prerequisite edges among tasks (e.g. A1-HAbst → A1-CORE; S1-S7 → S8).

Table 10. Hierarchical-workflow representation of the task tree (idempotent on the tree's name).

9. Open verification gaps

- **A2 force-reduction (highest priority).** The ~100 pN-nN catalytic force figure and selectivity-under-load are the program's load-bearing claims and are not yet machine-verified; A2-CORE's verifier (DFT cycle + AFM force spectroscopy) must return real numbers before Route A2 is admitted past hypothesis.
- **Energy bit-op basis.** B4 needs “bit-operation” defined per 8-bit addition before its ceiling is enforceable.
- **Cross-route stiffness ratification.** S1-S4 carry TBR stiffness/force numbers that resolve only once a route is chosen; both branches are kept live in the tree.
- **Reference-target library.** The judge-specified, test-time-revealed objects (and their pre-validated metrology) referenced by A1-CORE/A2-CORE/S8 remain to be specified — the natural next document.

References & notes

1. R. A. Freitas Jr. & R. C. Merkle, [“A Minimal Toolset for Positional Diamond Mechanosynthesis.”](#) J. Comput. Theor. Nanosci. 5, 760–861 (2008). 3 primary + 6 auxiliary tools (HAbst, HDon, DimerP; AdamRad, GeRad, Meth, GM, Germ, HTrans); elements C/Ge/H from feedstocks CH₄, C₂H₂, Ge₂H₆, H₂; 100% process closure; 65 reaction sequences, 328 steps, 354 pathological side reactions, 1,321 DFT energies over 1,620 tool/workpiece structures at B3LYP/6-311+G(2d,p)//B3LYP/3-21G*; high reliability at 80 K, moderate at 300 K.
2. K. E. Drexler, *Nanosystems: Molecular Machinery, Manufacturing, and Computation* (Wiley, 1992) — manipulator-arm stiffness, actuation, and positional-error analysis underlying the §6 subsystem targets.
3. Capability axes, the ½-uncertainty metrology rule, and the per-capability acceptance protocol: NDI Definition Set, Document 2, *Extending the Definitions of Molecular Nanotechnology* (2026), §2–3.
4. Stiffness arithmetic (force-delivery $k \approx 60$ N/m for ~6 nN within 0.1 nm; thermal floor $k \approx 0.41$ N/m): NDI Definition Set, Document 1, Appendix B.

— End of Document 2A —